THE STEIN FACTORIZATION OF A PROPER FLAT MORPHISM WITH GEOMETRICALLY REDUCED FIBRES IS FINITE ETALE

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Editorial note. Counted span is the author's own proof environment, source lines 8436-8476, immediately following the statement at 8425-8434.

Editorial note. The statement cites the Stein factorization theorem and the definition of geometrically reduced fibres by \ref; both are navigation pointers.

Editorial note. Two cross-chapter pointers (More on Morphisms, Derived Categories of Spaces) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Stein factorization in general. chapter theorem theorem-stein-factorization-general, cited in the statement. Supplies the factorisation X -> Y' -> Y whose second map is shown to be finite etale.
- Compatibility of the Stein factorization with flat base change. chapter lemma lemma-stein-universally-closed, cited at proof line 8438. Allows the fibre of Y' -> Y to be computed after a flat base change.
- Proper flat morphisms with geometrically reduced fibres have geometrically connected fibres nearby. chapter lemma lemma-proper-flat-geom-red, cited at proof lines 8468-8469. Upgrades connectivity of one geometric fibre to connectivity of the fibres in a neighbourhood.
- The residue field version of the Stein argument. chapter lemma lemma-stein-universally-closed-residue-fields, cited in the proof. Shows the fibre of Y' -> Y is a single point with the residue field of the base point.
- Structure sheaf of a proper flat morphism with geometrically connected fibres. Derived Categories of Spaces, lemma spaces-perfect-lemma-proper-flat-geom-red-connected, cited at proof lines 8473-8474. Gives f_*O_X = O_Y, which identifies Y' with Y locally and yields etaleness.

Source reference. Original definition 0E08: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6904–6904. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0E1E: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8387–8387. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0A19: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7994–7994. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0E1C: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8061–8061. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BSR: focused-seeds/geometry/S9-spaces-stein-factorization-etale/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0E0S: focused-seeds/geometry/S9-spaces-stein-factorization-etale/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 0A1C: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8197–8197. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a proper morphism of algebraic spaces over S. Let $X \to Y' \to Y$ be the Stein factorization of f (Theorem 0A1C). If f is of finite presentation, flat, with geometrically reduced fibres (Definition 0E08), then $Y' \to Y$ is finite étale.*

Proof. Formation of the Stein factorization commutes with flat base change, see Lemma 0A19. Thus we may work étale locally on Y and we may assume Y is an affine scheme. Then Y' is an affine scheme and $Y' \to Y$ is integral.

Let $y \in Y$. Set n be the number of connected components of the geometric fibre $X_{\overline{y}}$. Note that $n < \infty$ as the geometric fibre of $X \to Y$ at y is a proper algebraic space over a field, hence Noetherian, hence has a finite number of connected components. By Lemma 0E1C there are finitely many points $y'_1, \ldots, y'_m \in Y'$ lying over y and for each i we can pick a finite type point $x_i \in |X_y|$ mapping to y'_i the extension $\kappa(y'_i)/\kappa(y)$ is finite. Thus More on Morphisms, Lemma 0BSR tells us that after replacing Y by an étale neighbourhood of y we may assume $Y' = V_1 \amalg \ldots \amalg V_m$ as a scheme with $y'_i \in V_i$ and $\kappa(y'_i)/\kappa(y)$ purely inseparable. In this case the algebraic spaces $X_{y'_i}$ are geometrically connected over $\kappa(y)$, hence $m = n$. The algebraic spaces $X_i = (f')^{-1}(V_i)$, $i = 1, \ldots, n$ are proper, flat, of finite presentation, with geometrically reduced fibres over Y. It suffices to prove the lemma for each of the morphisms $X_i \to Y$. This reduces us to the case where $X_{\overline{y}}$ is connected.

Assume that $X_{\overline{y}}$ is connected. By Lemma 0E1E we see that $X \to Y$ has geometrically connected fibres in a neighbourhood of y. Thus we may assume the fibres of $X \to Y$ are geometrically connected. Then $f_*\mathcal{O}_X = \mathcal{O}_Y$ by Derived Categories of Spaces, Lemma 0E0S which finishes the proof. □